

Raman Kumar Singh

Pursuing Doctor of Philosophy (PhD)

Department of Energy Science and Engineering

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Education

Doctor of Philosophy (Ph.D.)

Department of Energy Science and Engineering, IIT Bombay, Mumbai, India

Supervisor: Prof. Prakash C. Ghosh, Prof. Venkatasailanathan Ramadesigan, & Prof. Amartya Mukhopadhyay **2021–Current**

Thesis: Modelling and Experimental Analysis of Metal–CO₂ Based Electrochemical Systems for CO₂ Utilization.

Master of Science (M.Sc.)

Department of Energy Science and Engineering, IIT Bombay, Mumbai, India

Supervisor: Prof. Prakash C. Ghosh & Prof. Venkatasailanathan Ramadesigan **2019–2021**

Thesis: A Modeling Approach for Na–Water Based Electrochemical System for CO₂ Capture.

Bachelor of Science (B.Sc.)

University of Allahabad, Prayagraj, India

2015–2018

Research & Professional Experience

- **Ph.D. Project:** Modeling and Experimental Analysis of Metal–CO₂ Based Electrochemical Systems for CO₂ Utilization

Supervisors: Prof. Prakash C. Ghosh, Prof. Venkatasailanathan Ramadesigan, Prof. Amartya Mukhopadhyay **July 2021–Present**

- **Mathematical Modeling of Mg–CO₂ System**

- **Aim:** Develop a 3D transient isothermal model of a membrane-free Mg–CO₂ primary battery for CO₂ utilization.
- **Methodology:** Integrated Euler–Euler multiphase modeling, dilute solution theory, and electrochemical kinetics with parametric analysis.
- **Outcome:** Enhanced power output and hydrogen production through optimized cathode porosity, improved electrolyte conductivity, and reduced anode–cathode spacing.

- **Multiphase Flow Modeling of Zn–CO₂ Battery**

- **Aim:** Develop a 2D transient Zn–CO₂ electrochemical model for CO₂ utilization and clean hydrogen production.
- **Methodology:** Integrated Euler–Euler two-phase flow, dilute solution theory, porous electrode modeling, and Darcy’s law, with experimental validation.
- **Outcome:** Improved system performance through enhanced cathode porosity, reaction kinetics, and reduced cathode–membrane spacing.

- **Flow-Based Zn–CO₂ Battery Modeling**

- **Aim:** Develop a scalable flow-based Zn–CO₂ battery model for continuous CO₂ utilization and energy generation.
- **Methodology:** Coupled electrolyte flow dynamics with electrochemical kinetics and conducted parametric analysis of operating and design variables.

- **Outcome:** Identified CO₂ pressure and tank volume as dominant performance drivers, while flow rate primarily influenced pumping losses.
- **Design and Modeling of Na–CO₂ Electrochemical System**
 - **Aim:** Develop a Na–CO₂ electrochemical system for integrated hydrogen generation, carbon utilization, and electricity production.
 - **Methodology:** Built and validated a multiphase transient model using dilute solution theory, binary electrolyte theory, and laminar bubble flow.
 - **Experiments & Materials:** Synthesized and characterized NASICON solid electrolyte, and designed an experimental Na–CO₂ cell to validate modeling predictions.
 - **Outcome:** Identified cathode–separator spacing, electrode porosity, anode size, and reaction kinetics as key optimization parameters.
- **Development and Optimization of Electrolyzer, Fuel Cell, and Zn–Air Battery Systems**
(Lab Collaboration / Supervisor: Prof. Prakash C. Ghosh) July 2020–Present
 - Involved in **electrolyzer stack** assembly and performance testing, electrochemical characterization of **Zn–air batteries**, and simulation-driven fuel cell performance analysis using **COMSOL**.
- **Modeling Approach for a Na–Water Electrochemical System for CO₂ Capture**
(M.Sc. Project / Supervisors: Prof. Prakash C. Ghosh, Prof. Venkatasailanathan Ramadesigan) July 2020–Dec 2021
 - Developed a Na–CO₂ electrochemical model for simultaneous **hydrogen production** and **electricity generation**, incorporating dilute solution theory, binary electrolyte theory, and laminar bubble flow. System performance was optimized by varying geometry, CO₂ flow rate, electrode spacing, and porosity.
- **Solid Oxide Fuel Cell Microstructure and Properties**
(M.Sc. Seminar / Guide: Prof. Dayadeep Moder) Jan 2020–May 2020
 - Analyzed three-phase boundary density, contact area per unit volume, hydraulic radius, and effective conductivity, focusing on microstructural parameters such as particle size, porosity, pore former radius, and contact angle.

Journal Publications

1. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. Mathematical modeling and simulation of Mg–CO₂ primary battery for hydrogen generation and CO₂ utilization. *Electrochimica Acta*, 146333 (2025). <https://doi.org/10.1016/j.electacta.2025.146333>
2. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. Multiphase Flow Modeling of an Aqueous Zn–CO₂ System for CO₂ Utilization and Hydrogen Production. (Submitted)
3. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. A flow model for simultaneous CO₂ conversion, H₂ production, and power generation in a Zn–CO₂ electrochemical system. (Submitted)
4. **Raman K. Singh**, D. Saha, S. Karmakar, P. C. Ghosh, A. Mukhopadhyay, & V. Ramadesigan. Modeling and experimental investigation of a Na–CO₂ system for hydrogen and power generation. (Manuscript in preparation)

Conference Proceedings

1. **Raman K. Singh**, D. Saha, S. Karmakar, P. C. Ghosh, A. Mukhopadhyay & V. Ramadesigan (**Poster Presentation**). Multifunctional Na–CO₂ electrochemical system: experimental validation and modeling for sustainable hydrogen and NaHCO₃ co-production. Presented at 13th Asian Conference on Electrochemical Power Sources (ACEPS 13), Bengaluru, India, January 11-14, 2026.

2. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan (**Oral Presentation**). Flow-based Zn–CO₂ battery for simultaneous CO₂ mitigation, power generation, and hydrogen production. Presented at 9th International Conference on New Energy and Applications (ICNEA 2025), Tokyo, Japan, December 28-30, 2025.
3. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan (**Poster Presentation**). Model-based investigation of Zn–CO₂ electrochemical systems for CO₂ reduction to H₂ and electricity production. Presented at SustainX - ACE 2025 & 4th CO2India Meet, IIT Madras, Chennai, India, June 21-24, 2025. **Best Poster Presentation Award**.
4. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan (**Poster Presentation**). A combined chemical electrochemical and multiphase flow Model of Zn–CO₂ system for CO₂ utilization in H₂ and electricity. Presented at ICOSSE'25, Auckland, New Zealand, February 20-22, 2025.
5. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan (**Poster Presentation**). Study on the effects of bubble diameter and reaction rate kinetics in an Mg–CO₂ electrochemical system. Presented at the EIHE 2025, Mumbai, India, January 21-25, 2025.
6. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan (**Poster Presentation**). A mathematical modeling approach to utilize CO₂ into hydrogen and electricity via Na–CO₂ electrochemical system. Presented at the Conference on Advances in Chemistry for Energy and Environment (CACEE), TIFR Mumbai, India, December 16-20, 2024.
7. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan (**Poster Presentation**). Modeling and numerical study of Mg–CO₂ electrochemical system for CO₂ utilization with electricity and hydrogen production. Presented at the 37th Topical Meeting of ISE, Stresa, Italy, June 8-12, 2024.
8. **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan (**Oral Presentation**). Electrical field effect on CO₂ absorption and chemisorption in a rectangular bubble column by COMSOL. **In Proceedings of the COMSOL Conference Bengaluru**, India, November 30 - December 1, 2023.

Workshops Participated

- SustainX – ACE 2025 & 4th CO2India Meet: Decoding Decarbonization & Accelerating Clean Energy Symposium, organized by the School of Sustainability, IIT Madras, India **June 21st–22nd, 2025**
- Green H₂ Technologies Workshop: Insights from Industry & Academia, organized by the Department of Energy Science and Engineering, IIT Bombay, India **Nov 8th, 2024**
- COMSOL Multiphysics Workshop, organized by the Department of Mechanical Engineering, IIT Bombay, India **Oct 23rd–24th, 2024**

Technical Skills & Instrumental Expertise

Software	COMSOL Multiphysics, MATLAB, Octave, OriginPro, Microsoft Office, L ^A T _E X, Biologic EC-Lab, AfterMath, ANSYS FLUENT, SolidWorks, AutoCAD, Design Builder
Programming	Python (NumPy, SciPy, Matplotlib), C, C++
Instruments	Biologic multichannel electrochemical workstation, Arbin battery cycler, Pine potentiostats, Vacuum oven, Muffle furnace, Hot press, UV-Vis spectroscopy, GC-TCD/FID
Characterization	CA, CP, CC, Tafel analysis, Polarization curves, GCD, CV, LSV, EIS, OCP, ECSA estimation, Faradaic efficiency analysis.

Academic Experience

Teaching Assistantship (Department of Energy Science and Engineering, IIT Bombay)

- Conservation of Energy in Buildings July 2024 – Dec 2024 & July 2025 – Dec 2025
- Non-Conventional Energy Systems Laboratory Jan 2025 – June 2025
- Fuel Cells Jan 2023 – June 2023
- Energy Laboratory Jan 2022 – June 2022

Social Responsibility

- Member of the Organizing Committee, 9th International Conference on Advances in Energy Research (ICAER 2023), IIT Bombay Dec 12th–14th, 2023
- Hostel Sports Secretary, Hostel 14, IIT Bombay 2022–2023
- Member, Institute Research Student Camping Programme (IRSCP), IIT Bombay — mentoring incoming students 2022–2023
- Department Sports Secretary, Department of Energy Science and Engineering, IIT Bombay 2021–2022
- Core Member, Organizing Committee, Energy Day Event, IIT Bombay May 15th, 2021

Academic Course Projects

- Estimated diffusion coefficients of ionic species using linear sweep voltammetry. Jan 2021 – Jun 2021
- Reviewed cathode materials and catalyst effects on Na–CO₂ battery performance, focusing on capacity, coulombic efficiency, and cycle stability. Jan 2021 – Jun 2021
- Assessed environmental impact of daily activities through carbon footprint analysis, with emphasis on energy and resource consumption. Jul 2020 – Dec 2020
- Simulated building cooling energy requirements (e.g., Shunya House) by optimizing material selection, glazing, orientation, and geographic parameters to minimize energy consumption. Jul 2019 – Nov 2019

Achievements & Extra-Curriculars

- Member, IIT Bombay Ultimate Frisbee Team (Night Crawlers) 2023–Present
- Sports Citation, Sports Color, and Cultural Special Mention Awards, Hostel 14 2023–2025
- **Best Secretary, Hostel 14** — Awarded for exceptional contributions to hostel activities and student engagement 2022–2023
- **Volunteering:** Actively contributed as a volunteer at multiple events, including Aahvan 2023 (IIT Bombay), IIT Bombay Half Marathon 2019, and student engagement activities 2019–2023
- **Sports Achievements:** Gold medalist in General Championship (GC) as a team member in Cricket and Ultimate Frisbee (2023–24), Silver medalist in Tug of War (2022–23), Bronze medalist in GC Football representing Tansa House (2019–20), Silver medalist in PGGC Cricket as part of the Energy Science team (2021–22), and Finalist in Hockey GC (2024–25).